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ENG EK301

Preliminary Design Report

November 16, 2025

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Introduction:

Analyzing truss structures requires solving a large system of equilibrium equations that relate member forces, reaction forces, and externally applied loads. For small structures this can be done by hand in a mostly straightforward manner, but as the number of joints and members increases, manual analysis becomes time-consuming, repetitive, and prone to error. In engineering practice, computational tools are essential for modeling structural systems efficiently and accurately. The purpose of this project is to develop a MATLAB program that automates the complete analysis of any statically determinate truss. The program builds the equilibrium matrix, computes internal member forces for an applied test load, evaluates buckling strength for each compression member, and determines the maximum load the structure can safely support. It also calculates the total cost of each design and the load-to-cost ratio, allowing objective comparison between alternative truss geometries.

Theory:

Truss analysis is based on modelling each element as a two-force member (hereby referred to as 2FM) that only carries axial loads (tension and compression) through friction-less pin joints. The truss geometry defines the direction cosines that can assist in decomposing the axial loads into their x and y components, defined as matrix A. Applying static equilibrium at each joint (method of joints) or splitting the structure into sections (method of sections) produces a system of 2J linear equations that relate all force members and support reactions to the externally applied loads L. These equations are assembled into a global equilibrium matrix, which is solved to the form

$$AT = L$$

to obtain the inner forces T. Members in compression may fail through buckling rather than material crushing. To determine the critical load, the truss will use the EK301 buckling strength model:

$$P_{crit} = 1909.7L^{-1.74} \pm 1.58oz$$

Where L is the length of the member in inches. The maximum load a truss can bear can modelled with the equation:

$$W_{failure} =- P_{crit}/R_{m}$$

Where R_m is the member's force per unit load. Finally, the cost model:

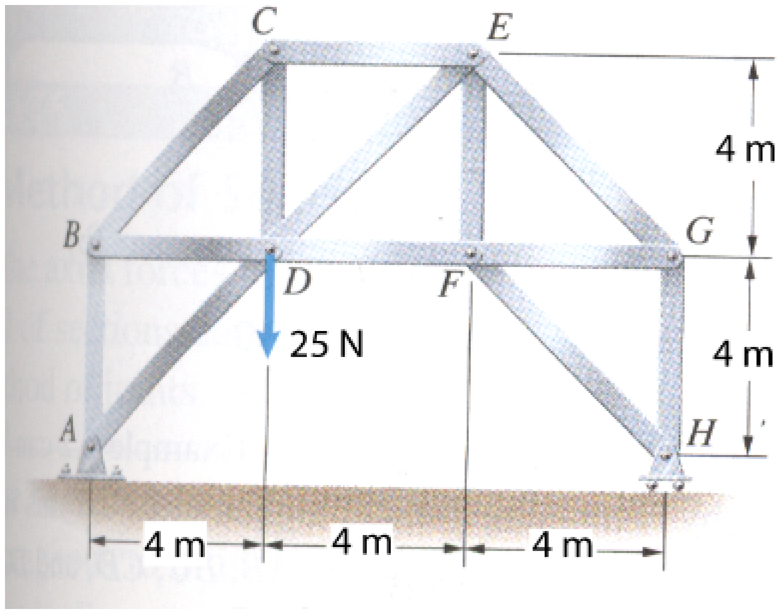
$$C = \$_{301}(L_{total} + 10J)$$

allows for a straightforward method of evaluating design efficiency through the load-cost ratio, which serves as the primary means for comparing truss geometries.

Analysis:

Model Verification:

Before analyzing any proposed designs, it is necessary to demonstrate that the MATLAB program produces accurate and reliable results. To verify correctness, one of the assigned practice trusses was analyzed both manually and computationally, and the resulting member forces were compared. The diagram of the verification truss is as follows:



Computational Approach

The system involved creating matrices described in the theory section, based on user imputed values. The model then solves those matrices for the forces on each member. The computational approach is as follows:

1. Acquire the number of joints J and members M
2. Create a connection matrix C with J rows and M columns, with two connected joints in each member column
3. Create X and Y vectors of the joint locations
4. Create an L vector of length 2J. The top J rows are x loads and the bottom J are y loads. Joint D is loaded by the 25N weight
5. Identify the joints providing support and create Sx and Sy matrices which identify the joints. Vertically concatenate the two matrices to create a S matrix with the top J rows as x supports and bottom J rows as y supports.

6. Create an A matrix with the x and y direction cosines for each member force in place of each joint in the C column and S horizontally concatenated to the right.
7. T will be a vector with the member forces and support forces at the bottom. Solve for T with A \ L. The program labels each force based on its sign for compression or tension.

With that computation and the values of the practice truss were inputted and the code gave:

```
EK301, Section A4, Group 1: Mark T., Mukund B., Silas L., 11/16/2025
Load: 25.00
Member forces in oz
m1: 16.667 (C)
m2: 23.570 (C)
m3: 16.667 (T)
m4: (ZFM)
m5: 16.667 (T)
m6: 16.667 (C)
m7: 11.785 (T)
m8: 8.333 (T)
m9: (ZFM)
m10: 11.785 (C)
m11: 8.333 (T)
m12: (ZFM)
m13: 8.333 (C)
Reaction forces in oz:
Sx1: 0.000
Sy1: 16.667
Sy2: 8.333
Cost of truss: 140.28$
```

Manual Approach

The calculations were performed with a combination of Method of Joints and Method of Sections, (Appendix A) and the results are as follows:

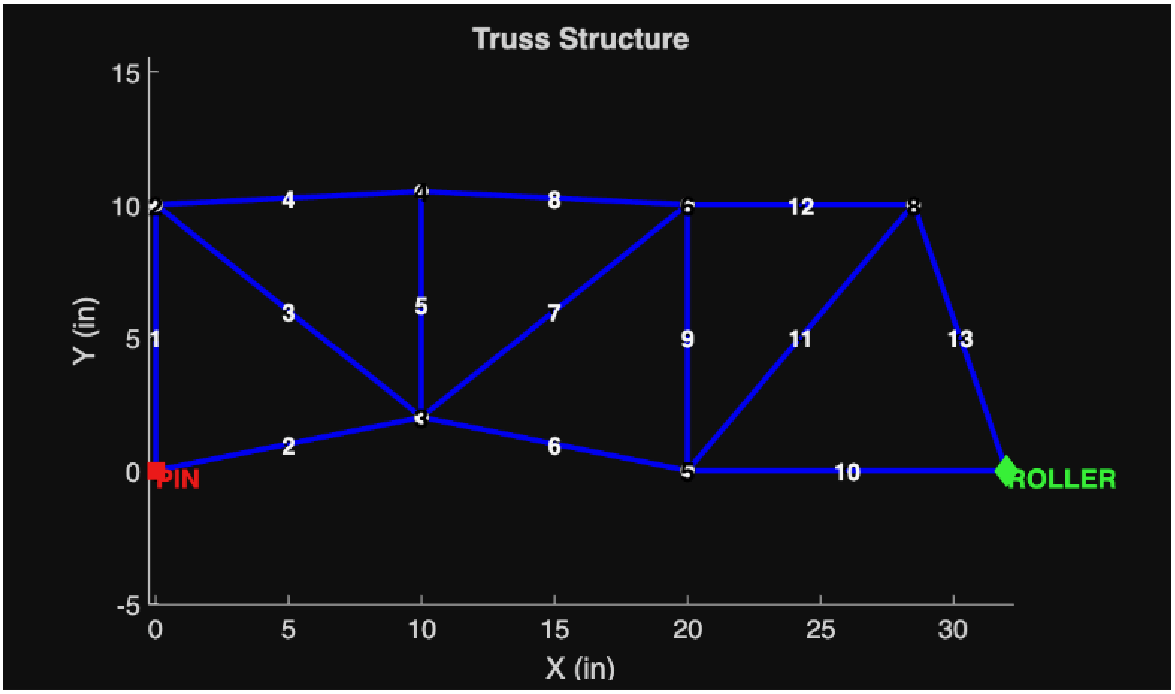
Member	Member Number	Force (N)	Tension or Compression
AB	m1	16.7	Compression
BC	m2	23.6	Compression
BD	m3	16.7	Tension
AD	m4	0	Zero force member
CD	m5	16.7	Tension
CE	m6	16.7	Compression
DE	m7	11.8	Tension
DF	m8	8.33	Tension
EF	m9	0	Zero force member
EG	m10	11.8	Compression
FG	m11	8.33	(Tension)
FH	m12	0	Zero force member
GH	m13	8.33	Compression
Ax	Sx	0	N/A
Ay	Sy_1	16.7	N/A
Hy	Sy_2	8.33	N/A

The consistency between the model and hand analysis demonstrates the validity of the program.

After this confirmation we moved onto our creation of truss designs.

Data and Designs:

Box Truss:



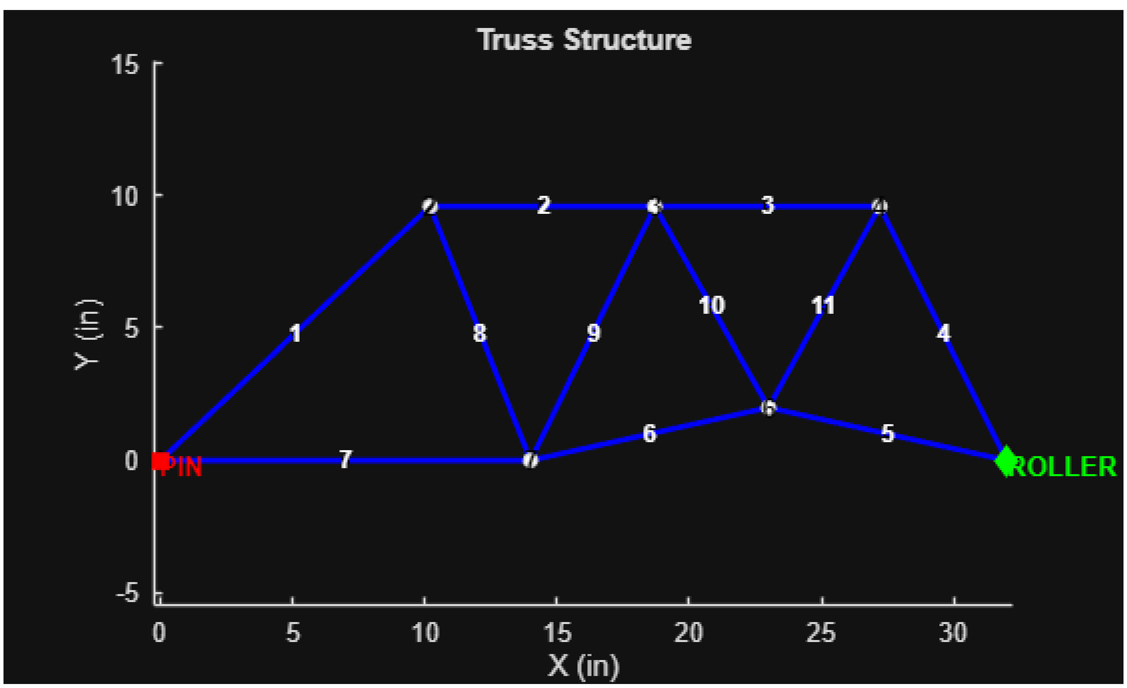
Member:	Force (oz)	Tension or Compression
m1	22.0	Compression
m2	0	Zero Force Member
m3	33.1	Tension
m4	25.9	Compression
m5	2.588	Tension
m6	12.2	Tension
m7	17.8	Tension
m8	25.9	Compression
m9	12.4	Compression
m10	3.5	Tension
m11	13.1	Tension

m12	12.0	Compression
m13	10.6	Compression

Supports	Force (oz)
Sx	0.000
Sy_1	22.0
Sy_2	10.0

Max Load:	42.8 ± 2.61oz
Total Cost:	$218.75
Unit Cost in max load/cost:	$0.196
Critical Member	m4

Rail Truss:



Member:	Force (oz)	Tension or Compression
m1	13.146	Compression
m2	13.146	Compression
m3	27.284	Compression
m4	28.931	Compression
m5	13.264	Tension
m6	20.189	Tension
m7	9.579	Tension
m8	9.685	Tension
m9	14.988	Compression
m10	15.381	Tension
m11	29.578	Tension

Supports	Force (oz)
Sx	0.000
Sy_1	9.004
Sy_2	22.996

Max Load:	34.042 ± 1.859
Total Cost:	$234.76
Unit Cost in max load/cost:	$0.186
Critical Member	m4

Sample Max Load Calculation in Matlab

```
%Finding the T vector
T=-(A \ L);

%Buckling force of each member
maxForceMatrix=zeros(M,1);
for m= 1:M
    bucklingF=(1909.713)*(memberLengthVector(m))^(-1.74);
    maxForceMatrix(m) = bucklingF;
end

%critical member
critRatio = 0;
for m = 1:M
    if T(m) < 0% compression only
        forceRatio = abs(T(m)) / maxForceMatrix(m);
        if forceRatio > critRatio
            critRatio = forceRatio;
            critMemberIndex = m;
        end
    end
end

%Finding Max Load
Rc=T(critMemberIndex)/abs(W);
pCrit=maxForceMatrix(critMemberIndex);
wFailure=pCrit/Rc;
```

To calculate the max load, first the buckling force was found for each member using the class fit data. Then, to find the critical member, the ratio of tension to maximum force was calculated, giving the weakest member experiencing the most load. The critical member was then used to find wFailure.

$$W_{failure} =- P_{crit}/R_m$$

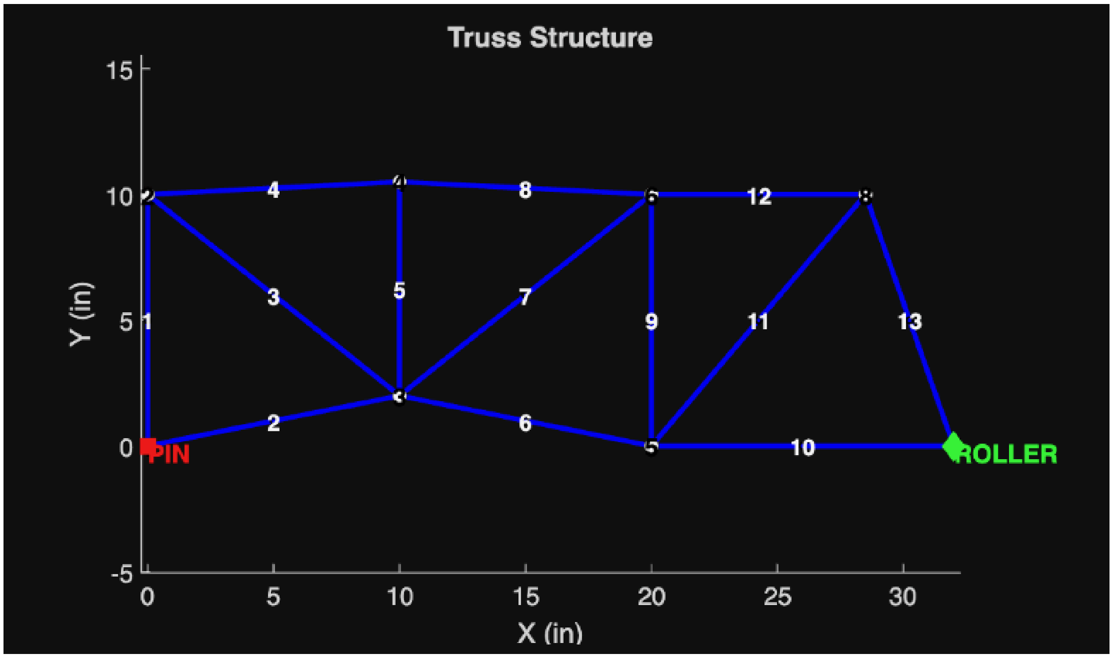
In earlier stages of the truss design, the critical member was found by determining the member experiencing the most compression. While this calculation would give the same results in our truss designs, the calculation was corrected because of its unreliability. The ratio method was used because even if a member experienced the most compression, it could have more compression strength compared to one that experiences only slightly less force.

Generative AI Usage

During the project AI was used exclusively in the coding of the analysis model. After losing practice with Matlab, ChatGPT was a valuable resource for finding specific functions and understanding how they worked together. AI code was never pasted directly into the analysis section and AI was never used to directly edit or create code. Instead AI was used as a reference to refresh ourselves on loops and solving equations in matlab. However, AI output was used directly in the creation of the graphing section of the design code. ChatGPT was prompted to create a graphing portion of the code using the given variables and created code that gave the blue Matlab truss graphs. This feature was extremely useful in speeding up our design process and especially in troubleshooting. Instead of having to manually find an incorrect connection between joints, one could look directly at the graph to see what the analysis code would be solving. We decided to directly use AI for this feature because it does not involve AI creating or editing our truss in any way, only improving our abilities to change it ourselves.

Results:

Box Truss:



Output:

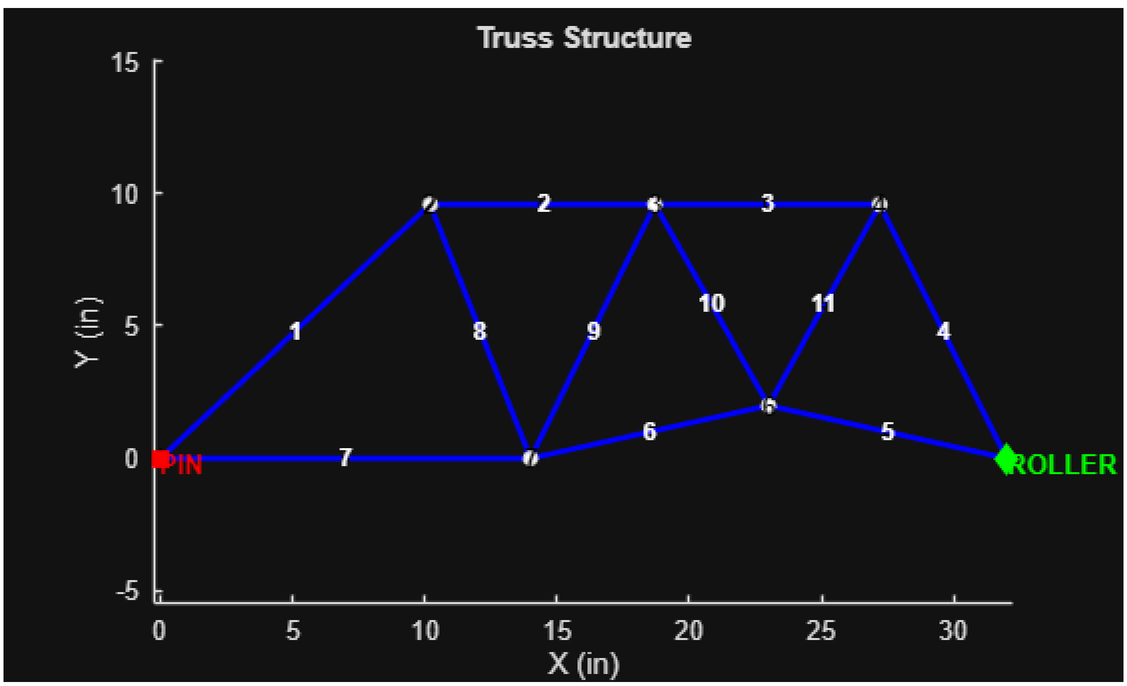
Member:	Force (oz)	Tension or Compression	Member Length (in.)
m1	22.0	Compression	10
m2	0	Zero Force Member	10.2
m3	33.1	Tension	12.8
m4	25.9	Compression	10
m5	2.588	Tension	8.5
m6	12.2	Tension	10.19
m7	17.8	Tension	12.8
m8	25.9	Compression	10
m9	12.4	Compression	10
m10	3.5	Tension	12
m11	13.1	Tension	13.1

m12	12.0	Compression	8.5
m13	10.6	Compression	10.6

Total Cost:	$218.75

The critical member was found to be member 4, which had a length of 10 in. and a buckling strength of 34.676 ± 1.58 oz. The max load was found to be 42.818 ± 2.611 oz. and the final cost/load ratio was calculated to be 0.196 oz./$.

Rail Truss:



Output:

Member:	Force (oz)	Tension or Compression	Member Length (in.)
m1	13.146	Compression	14

m2	13.146	Compression	8.5
m3	27.284	Compression	8.5
m4	28.931	Compression	10.7
m5	13.264	Tension	9.2
m6	20.189	Tension	9.2
m7	9.579	Tension	14
m8	9.685	Tension	10.3
m9	14.988	Compression	10.7
m10	15.381	Tension	8.7
m11	29.578	Tension	8.7

Total Cost:	$234.76

The critical member was found to be member 4, which had a length of 10.7 in. and a buckling strength of 28.931 ± 1.58 oz. The max load was found to be 34.042 ± 1.859 oz. and the final cost/load ratio was calculated to be 0.186 oz./$.

Discussion & Conclusion

Analyzing both the box truss and the rail truss, it was determined that the box truss was a better design due to its superior carrying capacity. The box truss design was able to carry a max load of 42.818 ± 2.611 oz while the rail truss design had a significantly lesser max load of 34.042 ± 1.859 oz, meaning that the box truss is capable of withstanding larger loads. Other factors went into the consideration of each design as well. The cost of the box truss was significantly lesser than the rail truss, with the box truss costing $218.75 and the rail truss having a final cost of $234.76. This makes the box truss much cheaper to produce than the rail truss, making it more cost effective alongside its superior carrying capacity.

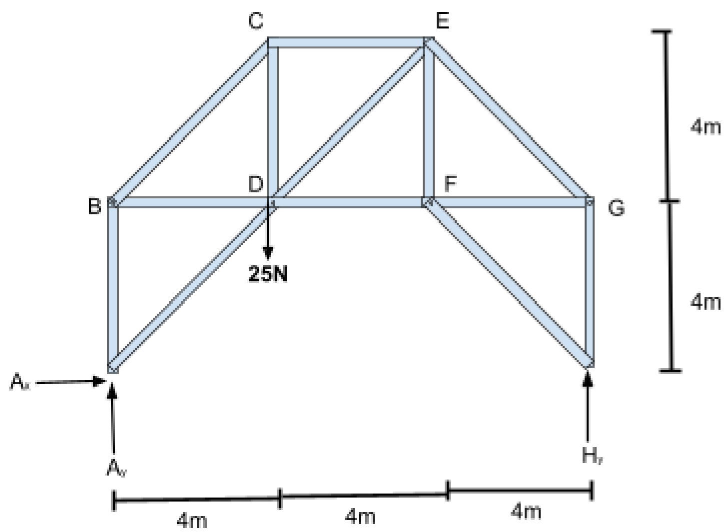
Other interesting facts found in each truss include the box truss' larger cost/load ratio of 0.196 oz./$ compared to the rail truss' 0.186 oz./$, further solidifying the cost effectiveness of the box truss. It was also found that the box truss design exerts more vertical force on the pin support than the roller support, ($Sy_1 = 22$ oz., $Sy_2 = 10$ oz.) while the rail truss exerts more force on the roller support than the pin support ($Sy_1 = 9.004$ oz., $Sy_2 = 22.996$ oz.). It was ultimately decided that this fact was irrelevant in the evaluation of the best design, however it was noted for any potential changes in the design later.

Improvements to this design include streamlining the lengths of each member. While the carrying capacity and cost is satisfactory, the lengths of certain members make it difficult to recreate in the real-world. Thus, it may be necessary to adjust the lengths of the members within a tolerance such that the carrying capacity is not lessened. The cost of the trust could also potentially be reduced by shrinking the overall height of the box truss by decreasing the length of certain members. Removing extraneous members could also be experimented with, which could also reduce the cost while preserving the carrying capacity.

APPENDIX

Appendix A: Hand calculations for verification truss

Full Body Analysis:



$$\Sigma F_X = 0 = A_x$$

$$\Sigma F_y = 0 = A_y + H_y - 25N$$

Ax = 0

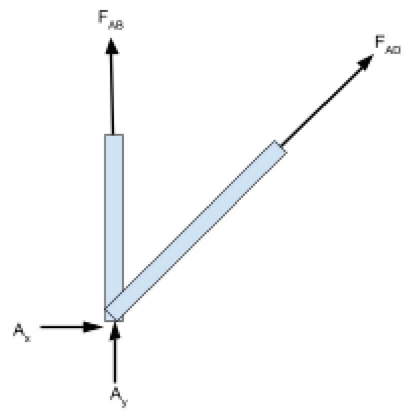
$$\Sigma M_A = 0 = -4m(25N) + 12(H_y)$$

Hy = 8.33N

Ay = 16.7N

Method of Joints:

1) **Joint A (Pin):**



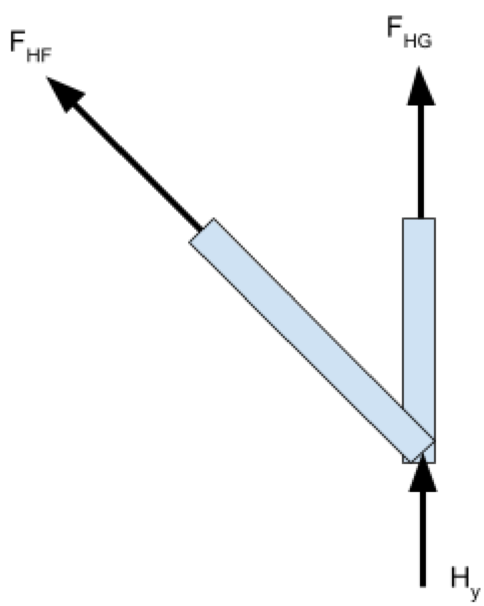
$$\Sigma F_x = 0 = F_{AD} cos(45)$$

F_{AD} = 0N (Zero force member)

$$\Sigma F_y = 0 = F_{AB} + A_y + F_{AD} sin(45) = F_{AB} + 16.7 + 0$$

F_{AB} = 16.7N (Compression)

2) **Joint H (Roller):**



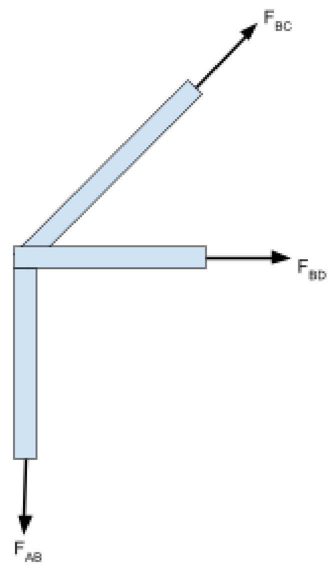
$$\Sigma F_x = 0 = F_{HF} cos(45)$$

F_{HF} = 0N (Zero force member)

$$\Sigma F_y = 0 = F_{HG} + H_y + F_{FH} sin(45) = F_{HG} + 8.33 + 0$$

F_{HG} = 8.33N (Compression)

3) Joint B



$$\Sigma F_x = 0 = F_{BD} + F_{BC} cos(45)$$

$$\Sigma F_y = F_{BA} + F_{BC} sin(45)$$

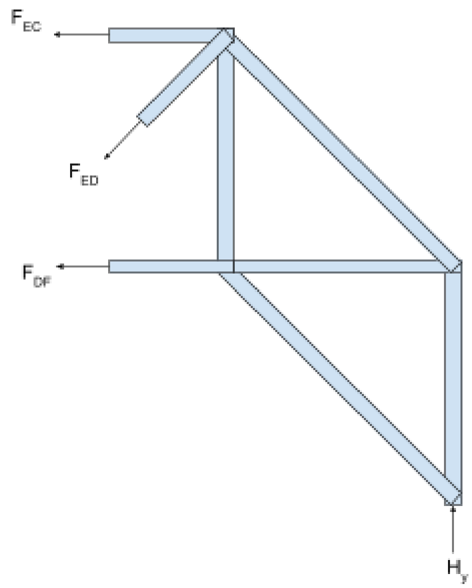
Recall that F_{AB} = +16.7N, so F_{BA} = -16.7

F_{BC} = 23.6N (Compression)

F_{BD} = 16.7N (Tension)

Method of Sections:

1) **FDB Right:**



$$\Sigma F_x = 0 = -F_{EC} - F_{DF} - F_{ed}cos(45)$$
$$\Sigma F_y = 0 = 8.33 - F_{ED}sin(45)$$

$\mathbf{F_{ED} = 11.8}$ **(Tension)**

$$\Sigma M_E = 8.33N(4m) - F_{DF}(4M) = 0$$

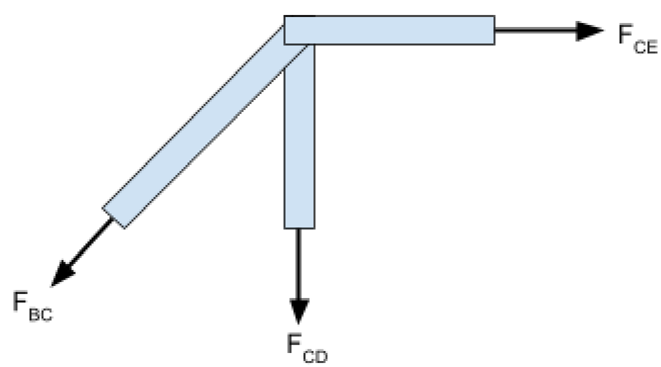
$\mathbf{F_{DF} = 8.33N}$ **(Tension)**

$$F_{EC} = -F_{DF} - F_{ED}cos(45)$$

$\mathbf{F_{EC} = 16.7N}$ **(Compression)**

Method of Joints (2nd time):

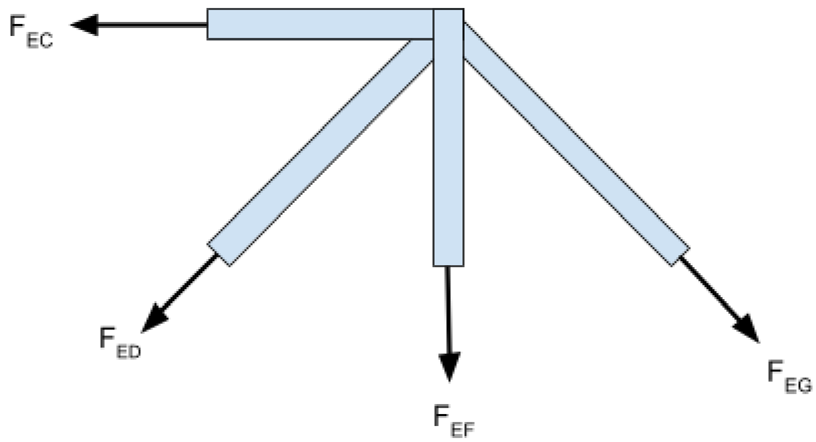
1) **Joint C:**



$$\Sigma F_y = 0 = -F_{CD} - 23.6sin(45)$$

$\mathbf{F_{CD} = 16.7N}$ **(Tension)**

2) Joint E:



$$\Sigma F_x = 0 = F_{EG}\left(\frac{4}{\sqrt{32}}\right) - F_{DE}\left(\frac{4}{\sqrt{32}}\right) - CE$$

(Plug in previously solved values)

F_{EG} = 11.8N (Compression)

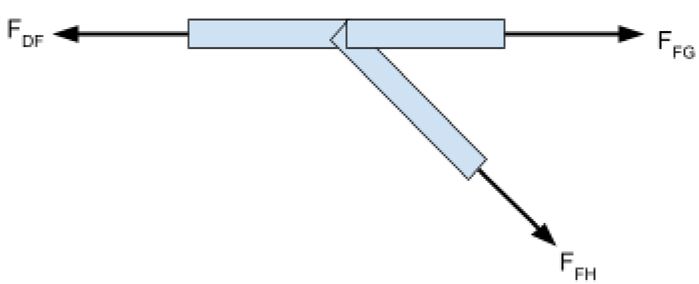
$$\Sigma F_y = 0 = - EF - EG\left(\frac{4}{\sqrt{32}}\right) - DE\left(\frac{4}{\sqrt{32}}\right)$$

(Plug in previously solved values)

F_{EF} = 0 (Zero Force Member)

3) Joint F:

Member EF is a Zero Force Member (no need to include in diagram)



F_{FH} = 0 (Zero Force Member)

F_{FG} = 8.33N (Tension)